Evaluation of Algorithms for Fractal Dimension Estimation in Medical Images: Complexity and Performance Analysis

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Resumo

Resumo do artigo em portugês

Palavras-chave: Palavras chave em português

Resumo

Abstract
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1 Introduction

Fractal geometry, formalised by Benoit B. Mandelbrot in his landmark monograph *The Fractal Geometry of Nature* (MANDELBROT, 1982), provided a mathematical framework for describing the roughness, branching, and statistical self-similarity pervasive in natural phenomena that classical Euclidean geometry fails to capture. The central quantitative descriptor of this framework is the fractal dimension D, estimated in practice as the negative slope of a linear regression over a log-log plot of a covering measure against the measurement scale ε. Shortly after Mandelbrot's unification, Pentland (PENTLAND, 1984) demonstrated that fractal dimension could be computed directly from digital imagery to discriminate surface textures in natural scenes — a result that bridged fractal geometry and image analysis, and opened the way for its systematic application to medical imaging tasks.

The adoption of fractal analysis in medical imaging occurs since the 1990s, alongside the proliferation of high-resolution digital modalities — multi-detector computed tomography, 3 T magnetic resonance imaging, and digital mammography. Applications spanning pulmonary emphysema (MISHIMA *et al.*, 1999), retinal vascular morphology (JELINEK; FERNANDEZ, 2000), brain cortical folding (FREE *et al.*, 1996), and tumour margin irregularity (RANGAYYAN; NGUYEN, 2007) demonstrated that D encodes structural properties that classical first-order statistical descriptors — variance, entropy, autocorrelation — could not adequately capture. From the 2010s onward, fractal descriptors were incorporated into hybrid feature engineering pipelines alongside learned representations (LOPES; BETROUNI, 2011; KING *et al.*, 2010), and more recently, fractal dimension has been explored as a a feature describer to enhance the interpretability of deep learning models (ROBERTO *et al.*, 2021a) and as a data augmentation strategy with GANs (BORGUE *et al.*, 2025a). Contemporary clinical datasets — commonly comprising images at resolutions exceeding 1024×1024 pixels — elevated the algorithmic efficiency of dimension estimation from a secondary concern to a primary design constraint. The central question shifted from whether fractal dimension is discriminative to at what computational cost it can be reliably extracted, and which algorithm offers the optimal balance of precision, robustness, and efficiency under realistic operating conditions.

Addressing this question requires a rigorous algorithmic engineering perspective. The four methods selected for analysis span the principal design axes of the problem: binary versus grayscale input, and non-overlapping versus dense-overlap covering strategies. The *Classical Box-Counting* (BC) (MANDELBROT, 1982) partitions the binary image into a grid of non-overlapping boxes of decreasing side and counts how many contain foreground pixels, yielding a coarse but efficient estimate of D. The *Gliding Box* (GB) (ALLAIN; CLOITRE, 1991) extends this idea by sliding a window across every pixel position, producing a richer mass-frequency distribution at the cost of processing a far larger number of box placements. The *Differential Box-Counting* (DBC) (SARKAR; CHAUDHURI, 1994) lifts the problem to grayscale images by treating the intensity surface as a three-dimensional relief and counting the column of boxes required to span the local intensity range within each cell, thereby retaining gradient information that binary methods discard. The *Blanket Method* (BM) (PELEG *et al.*, 1984) estimates the fractal dimension of the intensity surface by measuring how its apparent area grows as a morphological cover is progressively inflated and deflated around it at each scale. Contrasting these four methods also surfaces a fundamental trade-off inherent to the preprocessing stage: binarisation lowers the representational cost of the input, but discards intensity gradients that carry clinically significant discriminatory information — a loss that the grayscale methods avoid at the price of higher computational demand.

This work addresses the following research objectives:

- **Algorithmic analysis:** for each of the four methods, in both naive and structurally optimised formulations, derive formal asymptotic bounds for time and auxiliary space complexity and validate them against empirical measurements on the UCSB medical image dataset, contrasting theoretical scaling predictions with observed wall-clock behaviour.
- **Pre-processing and implementation cost:** quantify the overhead imposed by mandatory pre-processing stages and characterise the data-structure choices that drive the efficiency gap between naive and optimised implementations.
- **Effectiveness and trade-off synthesis:** assess the discriminatory power of each algorithm by measuring the statistical separation it induces between pathological and healthy tissue classes in terms of the estimated fractal dimension, and synthesise all time, memory, and effectiveness evidence into a structured trade-off comparison across the binary-versus-grayscale and naive-versus-optimised design axes.

The remainder of this article is structured as follows: Section 2 establishes the mathematical foundations of fractal analysis and formally defines each algorithm; Section 3 presents pseudocode and derives asymptotic complexity bounds; Section 4 describes the experimental methodology, dataset, and evaluation metrics; Section 5 reports and discusses empirical results; Section 6 concludes with recommendations and directions for future work.

2 Theoretical Foundation

2.1 Fractal Analysis and Dimension Estimation

Fractal geometry provides a mathematical framework for characterising structures whose complexity persists across observation scales, a property termed *statistical self-similarity* (MANDELBROT, 1982). Fractal sets exhibit power-law scaling: a characteristic measure $M(\varepsilon)$ obtained at scale ε satisfies

$$M(\varepsilon) \sim C \cdot \varepsilon^{-D}, \tag{1}$$

where C is a proportionality constant and D is the fractal dimension (FALCONER, 2003). Taking logarithms of both sides yields the linear relationship

$$\log M(\varepsilon) = \log C - D \log \varepsilon, \tag{2}$$

from which D is recovered as the negative slope of a least-squares regression over the log-scale pairs $\{(\log \varepsilon_i, \log M(\varepsilon_i))\}_{i=1}^{m}$, computed across m discrete scales.

Because the Hausdorff dimension (HAUSDORFF, 1919) requires an infimum over all possible coverings, it is not directly computable on discrete pixel grids. The Minkowski–Bouligand (box-counting) dimension D_B provides a tractable approximation (FALCONER, 2003):

$$D_B = \lim_{\varepsilon \to 0} \frac{\log N(\varepsilon)}{\log(1/\varepsilon)}, \tag{3}$$

where $N(\varepsilon)$ is the minimum number of boxes of side ε needed to cover the set. In practice, Eq. (3) is approximated by the finite regression of Eq. (2) over a scale set $\mathcal{S} = \{\varepsilon_1 > \varepsilon_2 > \cdots > \varepsilon_m\}$. The quality of the estimate is assessed through the coefficient of determination R^2 of the regression: values close to unity confirm consistent power-law scaling and are a necessary condition for interpreting D_B as a meaningful fractal descriptor (LOPES; BETROUNI, 2011).

2.2 Classical Box-Counting Method

The Classical Box-Counting (BC) method is the direct discretisation of Eq. (3) for binary images (MANDELBROT, 1982; LIEBOVITCH; TOTH, 1989). Let I be a binary image of $N = W \times H$ pixels. For a given box side ε, the image is partitioned into $\lceil W/\varepsilon \rceil \times \lceil H/\varepsilon \rceil$ non-overlapping cells; a cell is *occupied* if it contains at least one foreground pixel:

$$N(\varepsilon) = \sum_{i=0}^{\lceil W/\varepsilon \rceil - 1} \sum_{j=0}^{\lceil H/\varepsilon \rceil - 1} \mathbf{1}[\exists\,(x,y) \in B_{ij}(\varepsilon) : I(x,y) = 1]\,. \tag{4}$$

Because BC operates on binary images, a prior binarisation step is required when the input is a grayscale medical image. The standard strategy is Otsu's global thresholding (OTSU, 1979), which selects the threshold t^* that minimises intra-class intensity variance. This step introduces an inherent trade-off: binarisation lowers the computational cost of the subsequent coverage test, but irreversibly discards intensity gradient information, making the estimated D_B sensitive to the choice of t^*.

To avoid scanning all ε^2 pixels per cell, the optimised formulation pre-computes the *integral image* $P[x][y] = \sum_{i \le x, j \le y} I[i][j]$ using the recurrence (CROW, 1984)

$$P[x][y] = I[x][y] + P[x-1][y] + P[x][y-1] - P[x-1][y-1]. \tag{5}$$

Given P, the foreground pixel count within any rectangular cell is retrieved in $\mathcal{O}(1)$ as

$$\begin{aligned} \sigma(x_1, y_1, x_2, y_2) = P[x_2][y_2] - P[x_1-1][y_2] \\ - P[x_2][y_1-1] + P[x_1-1][y_1-1], \end{aligned} \tag{6}$$

with occupancy established by $\sigma > 0$. The integral image is built once in $\Theta(N)$ time and space, and shared across all scales in $\mathcal{S}$.

2.3 Gliding Box Method

The Gliding Box (GB) method (ALLAIN; CLOITRE, 1991) replaces the non-overlapping partition of BC with a *dense sliding window*: a box of side L is placed at every valid pixel position (x, y), and the *box mass*

$$m_{x,y}(L) = \sum_{i=0}^{L-1} \sum_{j=0}^{L-1} I[x+i][y+j] \tag{7}$$

is recorded for each of the $T(L) = (W - L + 1)(H - L + 1)$ valid placements. The resulting *mass-frequency distribution* $n(m, L)$ counts how many placements yield mass m, giving the normalised probability $P(m, L) = n(m, L)/T(L)$.

The fractal dimension is estimated from the mean box mass

$$\mu(L) = \sum_{m=0}^{L^2} m\, P(m, L), \tag{8}$$

by substituting $\mu(L)$ for $M(\varepsilon)$ in Eq. (2) and fitting the slope over $\mathcal{S}$ (PLOTNICK *et al.*, 1996). A complementary descriptor, *lacunarity*, is obtained from the first two moments of the distribution:

$$\Lambda(L) = \frac{\mu^{(2)}(L) - [\mu(L)]^2}{[\mu(L)]^2}, \tag{9}$$

where $\mu^{(2)}(L) = \sum_m m^2 P(m, L)$. Lacunarity quantifies translational heterogeneity: a homogeneous set yields $\Lambda \approx 0$, while a clustered or gapped structure yields large Λ (ALLAIN; CLOITRE, 1991; TOLLE; MCJUNKIN; GORSICH, 2008). The joint use of D_B and Λ has been shown to improve tissue discriminability in histological image classification (ROBERTO *et al.*, 2021b; BORGUE *et al.*, 2025b), since two textures may share a similar D_B yet differ in their spatial mass distribution.

The same integral-image structure of Eq. (5) applies to GB: pre-computing P reduces each mass evaluation from $\Theta(L^2)$ to $\mathcal{O}(1)$ via Eq. (6), bringing the per-scale cost from $\Theta(NL^2)$ to $\Theta(N)$ (CROW, 1984). This optimisation is more impactful for GB than for BC because GB evaluates $\Theta(N)$ overlapping positions per scale rather than $\Theta(N/L^2)$ disjoint cells, making the naive cost grow as L^2 while the optimised cost remains constant.

2.4 Differential Box-Counting Method

The Differential Box-Counting (DBC) method, proposed by Sarkar and Chaudhuri (SARKAR; CHAUDHURI, 1994), extends BC to grayscale images by treating the intensity surface as a three-dimensional relief. Let I be a grayscale image of $N = W \times H$ pixels with intensities in $\{0, \ldots, G-1\}$. For a given scale ε, the image plane is partitioned into $\lceil W/\varepsilon \rceil \times \lceil H/\varepsilon \rceil$ non-overlapping cells as in BC, but each cell is now extended vertically: the intensity axis is divided into bands of height $\delta = \lceil G/\lceil W/\varepsilon \rceil \rceil$, and the number of boxes required to cover the local intensity range within cell (i, j) is

$$n_{ij}(\varepsilon) = \left\lceil \frac{I_{\max}^{ij} - I_{\min}^{ij}}{\delta} \right\rceil + 1, \tag{10}$$

where $I_{\max}^{ij}$ and $I_{\min}^{ij}$ are the maximum and minimum intensities within the cell. The total count $N(\varepsilon) = \sum_{i,j} n_{ij}(\varepsilon)$ is substituted into Eq. (2) to estimate D_B via regression over $\mathcal{S}$.

By retaining gradient information through the intensity range $I_{\max}^{ij} - I_{\min}^{ij}$, DBC avoids the threshold sensitivity inherent to BC, at the cost of processing a richer input representation. The naive formulation scans all ε^2 pixels per cell to compute the local extrema, yielding a per-scale cost of $\Theta(N)$ with a constant proportional to ε^2. The optimised formulation replaces this scan with two separate *2D range-minimum and range-maximum structures* built in $\Theta(N \log N)$ time and $\Theta(N \log N)$ space via sparse tables, after which each $(I_{\max}^{ij}, I_{\min}^{ij})$ query is answered in $\mathcal{O}(1)$, reducing the per-scale traversal to $\Theta(N/\varepsilon^2)$ with a unit constant — the same asymptotic profile as the optimised BC applied to a grayscale input.

2.5 Blanket Method

The Blanket Method (BM), introduced by Peleg et al. (PELEG *et al.*, 1984), estimates the fractal dimension of the intensity surface by measuring how its apparent area grows as a morphological cover is progressively expanded around it at each scale. Let $u_\varepsilon(x, y)$ and $b_\varepsilon(x, y)$ denote the upper and lower *blankets* at dilation level ε, defined by the recurrences

$$u_\varepsilon(x, y) = \max\Big(u_{\varepsilon-1}(x, y) + 1,\ \max_{(s,t)\in\mathcal{N}} u_{\varepsilon-1}(s, t)\Big), \tag{11}$$

$$b_\varepsilon(x, y) = \min\Big(b_{\varepsilon-1}(x, y) - 1,\ \min_{(s,t)\in\mathcal{N}} b_{\varepsilon-1}(s, t)\Big), \tag{12}$$

with $u_0 = b_0 = I$ and $\mathcal{N}$ denoting the 4-connected neighbourhood of (x, y). The *blanket volume* at scale ε is

$$V(\varepsilon) = \sum_{x,y} \big[u_\varepsilon(x, y) - b_\varepsilon(x, y)\big], \tag{13}$$

and the scale-dependent area measure is obtained as $A(\varepsilon) = \big[V(\varepsilon) - V(\varepsilon-1)\big]/2$. Substituting $A(\varepsilon)$ for $M(\varepsilon)$ in Eq. (2) and fitting the slope over $\mathcal{S}$ yields the fractal dimension estimate.

The naive formulation evaluates each recurrence step by scanning the K-neighbourhood of every pixel, incurring $\Theta(NK^2)$ per dilation level and $\Theta(NK^2\varepsilon_{\max})$ in total. The optimised formulation decomposes the 2D neighbourhood maximum and minimum into two successive 1D sliding-window operations — one horizontal, one vertical — each solved in $\Theta(N)$ using a *monotonic deque* (double-ended queue) per row or column (LEMIRE, 2006). This separability reduces the per-level cost to $\Theta(N)$ regardless of neighbourhood size, and the total cost to $\Theta(N\varepsilon_{\max})$. Because BM operates directly on the intensity surface without binarisation, it retains the full gradient information of the image, making it the most information-preserving of the four methods at the price of higher computational demand.

3 Algorithms Analysis

4 Methodology

5 Results and Discussion

Resultados da pesquisa

Tabela 1: Legenda

col1	bol2	bol3	col4
dado1	dado2	dado3	dado4
dado1	dado2	dado3	dado4
dado1	dado2	dado3	dado4
dado1	dado2	dado3	dado4
dado1	dado2	dado3	dado4
resultado1	**resultado1**	**resultado1**	**resultado1**

nota de rodapé

6 Conclusion

Retomada da proposta, resultados e possíveis melhorias/trabalhos futuros.

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